Chapter 1: The Fundamental Unit of Life

While examining a thin slice of cork, Robert Hooke saw that the cork resembled the structure of a honeycomb consisting of many little compartments. Hooke called these boxes cells. A cell is the basic structural and functional unit of life. The plasma membrane is the outermost covering of the cell. It separates the contents of the cell from its external environment and permits the entry and exit of some materials. The cell membrane is therefore called a selectively permeable membrane. Diffusion is the spontaneous movement of a substance from a region of high concentration to a region where its concentration is low. Osmosis is the movement of water across a selectively permeable membrane.

Chapter 2: Tissues

A tissue is a group of cells that are similar in structure and work together to achieve a particular function. Plants have meristematic tissue and permanent tissue. Meristematic tissue is found at the growing tips of roots and stems and helps the plant grow. Permanent tissue includes simple tissue such as parenchyma, collenchyma and sclerenchyma, and complex tissue such as xylem and phloem. Xylem carries water and minerals upward from the roots. Phloem carries food from the leaves to other parts of the plant. Animal tissues include epithelial, connective, muscular and nervous tissue. Epithelial tissue covers the body and lines the internal organs.

Chapter 3: Diversity in Living Organisms

Living organisms show a wide range of diversity in size, shape, habitat and mode of nutrition. Classification groups organisms according to shared characteristics. The five kingdom system places organisms into Monera, Protista, Fungi, Plantae and Animalia. Monera includes bacteria, which are prokaryotic. Protista includes single celled eukaryotes such as Amoeba and Paramecium. Fungi include yeast, moulds and mushrooms. Plantae includes multicellular autotrophs. Animalia includes multicellular heterotrophs that usually move actively and ingest food.

Chapter 4: Motion

An object is said to be in motion if its position changes with time with respect to a reference point. Distance is the total length of the path travelled. Displacement is the shortest distance from the initial to the final position and has direction. Speed is distance divided by time. Velocity is displacement divided by time. Acceleration is the rate of change of velocity. A body moving with uniform velocity covers equal distances in equal intervals of time in a straight line. Graphs of distance against time help us compare rest, uniform motion and non-uniform motion.

Chapter 5: Force and Laws of Motion

Force is a push or a pull that can change the speed, direction or shape of an object. Newton's first law states that an object remains at rest or in uniform motion in a straight line unless acted upon by an unbalanced force. Inertia is the tendency of an object to resist a change in its state of motion. Newton's second law states that the rate of change of momentum is proportional to the applied force and takes place in the direction of the force. Newton's third law states that for every action there is an equal and opposite reaction.

Chapter 6: Gravitation

Every object in the universe attracts every other object with a force called gravitational force. The force is directly proportional to the product of the masses and inversely proportional to the square of the distance between them. The acceleration due to gravity near the surface of the Earth is approximately nine point eight metres per second squared. Weight is the force with which the Earth attracts an object. Mass is the amount of matter in an object and does not change with place, while weight changes with the value of gravity. Thrust and pressure are related to force acting on an area.